Year 9 Algebra Revision — Combining the Skills

Each question now uses more than one skill. Show every step of your working.

1 Simplify, then Solve

Collect like terms, then solve

1. $3x + 2x - 1 = 14$ _____
2. $5x - 3 - x = 13$ _____
3. $2x + 3x + 4 = 24$ _____

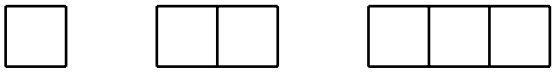
Expand the bracket, then solve

4. $2(x + 3) = 16$ _____
5. $4(x - 2) = 8$ _____
6. $3(x + 1) = 21$ _____

2 From a Pattern to a Line

For each pattern: look at the diagrams, complete the table, write the rule, plot the points and draw the line, then state the gradient and y-intercept.

Squares made from matchsticks

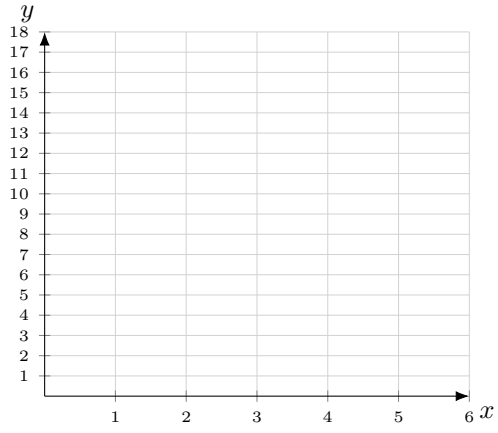


$n = 1$ $n = 2$ $n = 3$

n (squares)	1	2	3	4	5
matchsticks	4	7	10		

How many matchsticks are added each time a new square joins the row? _____

Rule: matchsticks = _____



Gradient: _____ y-intercept: _____

Triangles made from matchsticks

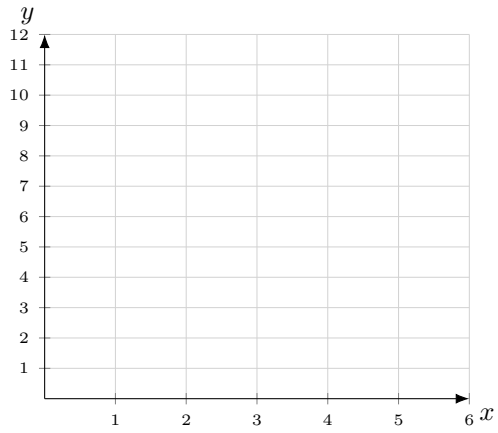


$n = 1$ $n = 2$ $n = 3$

n (triangles)	1	2	3	4	5
matchsticks	3	5	7		

How many matchsticks are added each time a new triangle joins the row? _____

Rule: matchsticks = _____



Gradient: _____ y-intercept: _____

3 Using a Line's Equation

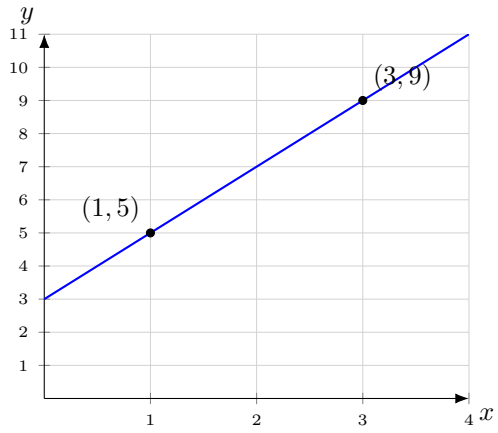
Use substitution for part (a), then solve an equation for part (b).

1. A line has equation $y = 2x + 5$.
   (a) Find y when $x = 3$. _____
   (b) Find x when $y = 17$. _____

2. A line has equation $y = -3x + 4$.
   (a) Find y when $x = -2$. _____
   (b) Find x when $y = -11$. _____

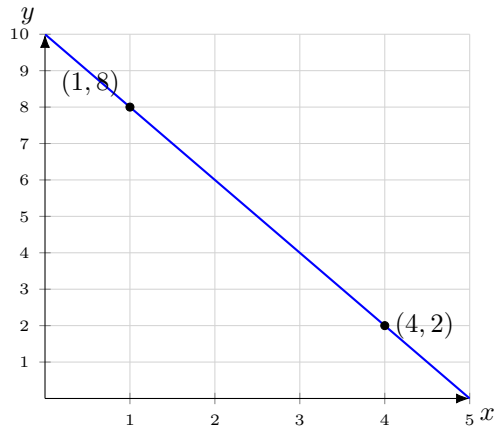
4 Finding the Equation from Two Points

Each graph shows two points on a line, but the y-intercept is not marked. Find the gradient from the two points, then substitute one point into $y = mx + c$ to find c.



Gradient: _____

Equation: _____



Gradient: _____

Equation: _____

5 Gradient and Equation from a Table

For each table, find the gradient and y-intercept, then write the equation in the form $y = mx + c$.

x	0	1	2	3
y	1	4	7	10

Gradient: _____

y-intercept: _____

Equation: _____

x	0	1	2	3
y	8	5	2	-1

Gradient: _____

y-intercept: _____

Equation: _____